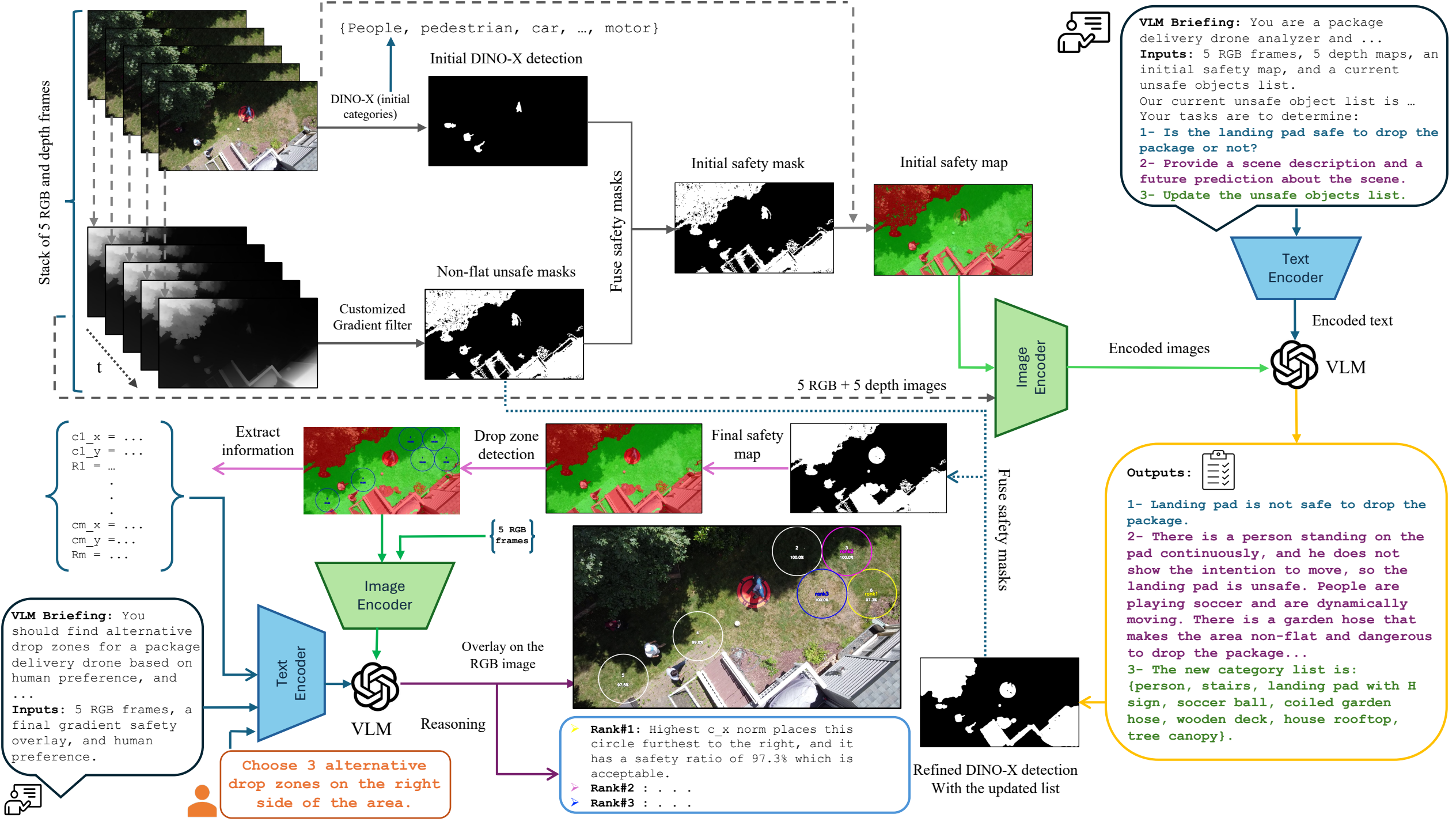
Stack of 5 RGB and depth frames
t
{People, pedestrian, car, …, motor}
DINO-X (initial categories)
Initial DINO-X detection
Customized Gradient filter
Non-flat unsafe masks
Fuse safety masks
Initial safety mask
Initial safety map
VLM Briefing: You are a package delivery drone analyzer and ...
Inputs: 5 RGB frames, 5 depth maps, an initial safety map, and a current unsafe objects list.
Our current unsafe object list is …
Your tasks are to determine:
1- Is the landing pad safe to drop the package or not?
2- Provide a scene description and a future prediction about the scene.
3- Update the unsafe objects list.
Text Encoder
Encoded text
Image Encoder
Encoded images
VLM
5 RGB + 5 depth images
Fuse safety masks
Final safety map
Drop zone detection
Extract information
c1_x = ...
c1_y = ...
R1 = …
cm_x = ...
cm_y =...
Rm = ...
5 RGB frames
Image Encoder
Text Encoder
VLM
Reasoning
Overlay on the RGB image
VLM Briefing: You should find alternative drop zones for a package delivery drone based on human preference, and ...
Inputs: 5 RGB frames, a final gradient safety overlay, and human preference.
Choose 3 alternative drop zones on the right side of the area.
Rank#1: Highest c_x norm places this circle furthest to the right, and it has a safety ratio of 97.3% which is acceptable.
Rank#2 : . . .
Rank#3 : . . .
Refined DINO-X detection
With the updated list
Outputs:
1- Landing pad is not safe to drop the package.
2- There is a person standing on the pad continuously, and he does not show the intention to move, so the landing pad is unsafe. People are playing soccer and are dynamically moving. There is a garden hose that makes the area non-flat and dangerous to drop the package...
3- The new category list is: {person, stairs, landing pad with H sign, soccer ball, coiled garden hose, wooden deck, house rooftop, tree canopy}.